

Differentiation Rules

LLE – Mathematics and Statistics

Power Rule

Given the function $y = ax^n$, the derivative is:

$$\frac{dy}{dx} = nax^{n-1}$$

Use this rule to differentiate:

1. $y = 5x^4$

2. $y = 7x^2$

3. $y = \frac{2}{3}x^6$

These indices facts will be useful for the next questions:

$$x^{-n} = \frac{1}{x^n}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

$$x^a \times x^b = x^{a+b}$$

$$\frac{x^a}{x^b} = x^{a-b}$$

$$(x^a)^b = x^{ab}$$

Use the power rule to differentiate:

4. $y = \frac{3x^3}{x^5}$

5. $y = \frac{10}{x^2}$

6. $y = 2x^{\frac{2}{3}}$

7. $y = 4\sqrt{x}$

8. $y = \frac{2}{3x^2}$

9. $y = \frac{10}{x}$

The power rule can be used to show:

$$y = ax, \quad \frac{dy}{dx} = a$$

$$y = a, \quad \frac{dy}{dx} = 0$$

Put this altogether and differentiate:

10. $y = 5x^2 + 4x + 3$

11. $y = x^4 - 2x^{-5} + \frac{1}{x}$

12. $y = 1 + 2\sqrt{x} + \frac{2}{5x} + \frac{3}{4x^2} + \frac{5}{2x^3}$

Exponentials

Given the function $y = ae^x$, the derivative is:

$$\frac{dy}{dx} = ae^x$$

Not many questions here, differentiate:

1. $y = 5e^x$

Logarithms

Given the function $y = a \ln x$, the derivative is:

$$\frac{dy}{dx} = \frac{a}{x}$$

Differentiate:

1. $y = 5 \ln x$

$$2. y = \frac{\ln x}{2} - 3e^x$$

The following log laws will be useful:

$$\ln(AB) = \ln A + \ln B$$

$$\ln\left(\frac{A}{B}\right) = \ln A - \ln B$$

$$\ln(A^n) = n \ln A$$

Differentiate:

$$3. y = \ln 4x$$

$$4. y = \ln 0.443x$$

$$5. y = \ln \frac{2}{x}$$

$$6. y = \ln x^{10}$$

$$7. y = 6 \ln \frac{1}{2}x$$

Mixed Questions

For each question find the requested derivative:

$$1. y = 5x^3 + x^2 - 4x + 1 + 3e^x + \ln 2x \quad \frac{dy}{dx} =$$

$$2. y = 2x + 5\sqrt{x} - \frac{4}{x} + \frac{3}{x^2} \quad \frac{dy}{dx} =$$

$$3. y = 5t^2 - 2t^{\frac{3}{4}} + 6 \ln 0.1t \quad \frac{dy}{dt} =$$

$$4. P = 6e^n - 4 + \frac{5}{2}n - \frac{6}{\sqrt{n}} \quad \frac{dP}{dn} =$$

$$5. G = 5s^2(3s - 4) \quad \frac{dG}{ds} =$$

By substituting into the function and into the derived function, find the value of the function and the value of the gradient at the given x values:

$$6. y = 5x^2 - 12x + 2, \quad x = 2$$

$$7. y = 4x^3 + x + 10, \quad x = -1$$

$$8. y = x^2 + 5x - 1, \quad x = -2.5$$

$$9. y = 4e^x - 10\sqrt{x}, \quad x = 2$$

$$10. y = 5 - \frac{5}{x} + \frac{5}{x^2}, \quad x = 1$$

Optimisation Extension (Partial Differentiation)

A consumer aims to maximize their utility from consuming two goods, x and y . Their utility function is given by:

$$U(x, y) = x^{0.8}y^{0.2}$$

The prices of the goods are $P_x = 4$ and $P_y = 2$, and the consumer has an income of $I = 200$.

Part of a solution:

Budget Constraint:

$$4x + 2y = 200$$

Lagrangian:

$$L(x, y, \lambda) = x^{0.8}y^{0.2} - \lambda(200 - 4x - 2y)$$

1. The Lagrangian function is provided above.

First Order Conditions:

Differentiate the Lagrangian with respect to x , y , and λ , and set each derivative to zero. Remember to apply the power rule for differentiation.

$$\frac{\partial L}{\partial x} = (\text{Differentiate } L \text{ with respect to } x, \text{ holding } y \text{ and } \lambda \text{ constant.}) = 0$$

$$\frac{\partial L}{\partial y} = (\text{Differentiate } L \text{ with respect to } y, \text{ holding } x \text{ and } \lambda \text{ constant.}) = 0$$

$$\frac{\partial L}{\partial \lambda} = (\text{Differentiate } L \text{ with respect to } \lambda, \text{ holding } x \text{ and } y \text{ constant.}) = 0$$

2. Write down the full first-order conditions, performing the differentiation steps.

Solving for Optimal Quantities:

3. You can rearrange $\frac{\partial L}{\partial x}$ to get: $0.8x^{-0.2}y^{0.2} = 4\lambda$. Now, create a similar rearranged expression for $\frac{\partial L}{\partial y}$.

Dividing the Equations:

You then divide these two rearranged equations by one another:

$$\frac{0.8x^{-0.2}y^{0.2}}{\boxed{\phantom{0.8x^{-0.2}y^{0.2}}}} = \frac{4\lambda}{\boxed{}\lambda}$$

4. Substitute your rearranged expression for $\frac{\partial L}{\partial y}$ from step 3 into the ratio above.

5. Simplify the resulting equation to express y in terms of x .

6. Substitute your equation for y into the budget constraint ($4x + 2y = 200$).

7. Solve the resulting equation for x .

8. Use the optimal x value to find the optimal quantity of y .

Extension Question

A consumer aims to maximize their utility from consuming two goods, x and y . Their utility function is given by:

$$U(x, y) = 2 \ln x + 3 \ln y$$

The prices of the goods are $P_x = 5$ and $P_y = 10$, and the consumer has an income of $I = 100$.